

TEST ONLY - SYNTHETIC ASSET DATA

Equipment Register and Data Sheets

Project CW-01 | P-101, VFD-101, PT-101 | Revision 1

This fixture validates document ingestion, retrieval, citations, OCR, and safe failure behavior. It is not an approved plant record and must not be used to operate or maintain equipment.

1. Equipment register

Tag	Equipment	Synthetic details
P-101	Vertical multistage centrifugal pump	42 m3/h at 55 m head; 15 kW; 2950 rpm
VFD-101	General-purpose variable-frequency drive	400 V class; 15 kW; PID control
PT-101	Discharge pressure transmitter	0-10 bar range; 4-20 mA output

2. Tag and relationship rules

- PT-101 is installed on the discharge header and supplies feedback to VFD-101.
- VFD-101 commands P-101 speed in the synthetic automatic-pressure test mode.
- Project membership does not grant mutation access to an Equipment record.
- Each Equipment source is linked once as a logical document; the Project inherits current versions.

3. P-101 data sheet

Attribute	Test value
Service	Clean cooling-water booster, synthetic
Nominal flow	42 m3/h
Nominal differential head	55 m
Motor	15 kW, 400 V class, 3 phase
Nominal speed	2950 rpm
Seal	Cartridge mechanical seal, synthetic descriptor
Observation points	Suction/discharge pressure, leakage, sound, vibration, motor current

The listed values are search and citation fixtures. They are not verified against a manufacturer nameplate.

4. VFD-101 and PT-101 data sheet

Attribute	VFD-101	PT-101
Function	P-101 speed control	Discharge pressure feedback
Signal	Receives 4-20 mA	Outputs 4-20 mA
Synthetic range	15 kW class	0-10 bar(g)
Project target	Maintains 4.2 bar(g)	Measures the controlled variable
Access note	Electrical work not authorized	Calibration not certified

5. Source applicability

The Equipment Register establishes project identity and synthetic data. The Pump Inspection Field Guide adds test observations for P-101. The Multimodal Control Loop Diagnostic describes written control-loop relationships while preserving one visual-only negative test.

6. Inspection record fields

Field	Format	Example only
Timestamp	ISO date/time with timezone	2026-08-14 09:40 IST
Pressure	bar(g)	3.1
Alarm/observation code	Controlled text	CW-17
Leakage	None / observed / not checked	None observed
Vibration	Measured value or qualitative observation	No numeric value on OCR card
Final wording	Human-reviewed text	No physical maintenance was performed